27 October 2026 | Discussion draft v2 | Not for approval of expenditure
MEMORANDUM

Panda Vision: Queensland model-based economic evaluation

Action required

Note the revised model and its material uncertainty; support use of existing evidence and explicitly reasoned estimates for local and Queensland investment decisions. Do not treat the current outputs as demonstrated savings or procurement approval. No new clinical study or structured elicitation is a prerequisite.

Author	Prepared for Dr Dylan Mordaunt, Chief Medical Officer
Presented by	Chief Medical Officer / nominated project lead

Executive summary

Purpose and scope. Compare video-supported with telephone-supported neonatal consultation. Both arms retain specialist consultation and the usual RSQ/ANTS-NQ escalation pathway. The existing pilot covers Atherton, Mareeba and Innisfail; 4–5 episodes occurred at Atherton. Expected activity is 10–15 consultations per site annually. The $10,000 installation estimate remains subject to invoice/component verification. [ELC2026; USER2026]

Approach. Use an annual-cohort model with low-, moderate- and critical-acuity pathways. Critical babies remain eligible for transfer; video failure returns to telephone. Include readiness/training, incremental video workflow, delayed rescue, local and receiving-hospital care, family burden, shared specialist overhead and stepwise capacity costs. Do not equate fewer transfers with better care. [M1]

Population. Public 2022 perinatal data identify 157 births at Atherton, 134 at Mareeba and 195 at Innisfail. The statewide roster preserves 46,711 public births across 65 co-located sites. A primary expansion scenario includes 33 non-hub sites with at least 20 births (28,276 births). These are historical denominators, not a verified 2026 capability list. Private-facility detail, reopened services and emergency-only birth locations remain distinct scope issues. [QH2022; ANZNN2023]

Evidence. The observational transfer OR of 0.48 is attenuated toward no effect and widened for transportability. Published economic models are structural comparators, not independent clinical replications. A 2026 simulation found no performance difference between phone and video support; therefore null and sceptical effect structures are explicit. No mortality benefit is assumed in the primary analysis. [HAY2020; ARM2012; YOO2022; THAO2022; LIETZ2026]

Costs and funding. A public Queensland aeromedical anchor of $18,670 is indexed to about $23,825 at the June 2026 price point, with 2019 treated as an uncertain source-price year. It is not a neonatal tariff. Geographic transport mixes and avoidable cost fractions are scenario assumptions. Keep resource costs, NWAU × NEP and QWAU × QEP separate. NEP26 is $7,418; current QEP/local weights are not verified and remain blank, not substituted. [AIR2019; CPIBASE; CPI2026; QHFUND26; NEP26]

Results. The three-site deterministic scenario costs $14,052 annually and gives $4,167 net resource savings. The 20,000-draw PSA gives 46% probability of cost saving, mean $2,428 and 95% interval –$41,148 to +$65,642. The 33-site illustrative state scenario gives 50.7%, mean $135,429 and interval –$1.012m to +$1.809m. These are assumption-dependent distributions, not an investment verdict. Shared effect uncertainty does not disappear as the network grows. [M1]

Analyses. Fifty-eight deterministic scenarios cover costs, equipment, geography, uptake, reliability, rescue, casemix and shared capacity. PSA separates parameter uncertainty from year-to-year random activity. An additional 108-point exploratory frontier includes extra survival with major morbidity and additional lifetime costs. The latter is not a calibrated cost-utility model. [M1]

Next decision. Use the model to identify robust portfolios, acceptable readiness costs and the benefit needed to justify investment. Ordinary invoices, available routine activity and public-source updates can replace assumptions when accessible; formal clinician estimation is not required. Procurement, privacy, biomedical approval and data-access authorisation remain separate. [M1]

Engagement and consultation	
Consumers	Family time, separation and access are included as consequences; no consumer-derived utility is claimed.
Clinicians and staff	Local information from Dr Nicholas Cairns and the ELC paper. Expert review is optional, not an evaluation dependency.
External stakeholders	Queensland Health, RSQ/ANTS-NQ, neonatal network and finance sources inform the model; no external endorsement is implied.

Implications	
Budget	No new expenditure approval sought. Distinguish $30,000 indicative replacement/establishment capital from sunk pilot capital. Annual continuation cash estimate $7,123; it is not a full economic cost.
FTE	No new funded establishment assumed. State scenarios include shared support and capacity increments; staff rates and time remain estimated.
Risk	Sparse evidence, source transportability, historical denominators, inferred geography and uncertain marginal costs. Null/harm scenarios and shared uncertainty prevent false precision.
Legal	Existing-data modelling only. Confirm governance and access before patient-level linkage or publication. No patient-level data are required for this version.
ICT	Tablet/PTZ replacement remains an unverified engineering option. Confirm QH Portal far-end control, power, mounting, cleaning and cyber/biomedical requirements.
Impact on First People's Health	Assess remote-access and family-displacement consequences separately. No cost-only rule or disability-based exclusion is proposed.

Strategic Plan alignment

Our care: timely specialist support. Our people: rural readiness. First Peoples health: equitable access and less displacement. Our sustainability: explicit resource, funding and uncertainty analysis.

Attachment/s

1. Economic model v2 workbook and reproducible outputs.
2. CHHHS decision presentation v2.
3. Methods/evidence supplement v2 (source identifiers above).
4. Verification report and Conductor project history.